Chapter 5

Pain Control for Operative Dentistry

"For there was never yet a philosopher, who could endure the toothache patiently".

—William Shakespeare, Much Ado About Nothing, Act V

Pain Control

Historically, the public has associated dental treatment with pain. This association is no longer valid because techniques for the elimination of pain, including atraumatic injection, have been available for years and are essential to a successful dental practice. Local anaesthesia for operative dentistry must be profound, often to depths required for pulpal anaesthesia. The following information, if understood and practiced, should eliminate pain associated with dental procedures. For additional information the reader is referred to *Malamed's Handbook of Local Anesthesia*.[1]

Local Anaesthesia

The administration of local anaesthesia to all tissues in the operating site is recommended in certain patients to eliminate pain and reduce salivation associated with tooth preparation and restoration. To administer effective anaesthesia, the dentist must have a thorough knowledge of the patient's physical and emotional status and an understanding of the effects of the drug to be injected and the advantages and disadvantages of adding vasoconstrictors.

Clinical Notes

- A *therapeutic dose* of a drug is the smallest amount that is effective when properly administered and does not cause adverse reactions.
- An *overdose* of a drug is an excessive amount that results in an overly elevated local accumulation or blood level of the drug, which causes adverse reactions.
- The normal healthy patient can safely receive *5–8 cartridges of anesthetic per appointment*. Each 1.8 ml cartridge contains anesthetic, with or without a vasoconstrictor (e.g. lidocaine 2% [anesthetic] with epinephrine 1:100,000 [vasoconstrictor], lidocaine 2% plain [no vasoconstrictor]).
- The number of permissible cartridges *increases as body weight increases*. According to Malamed, the maximum recommended dose of 2% lidocaine with epinephrine 1: 100,000 is *4.4* mg/kg, to an absolute maximum of 300 mg. Table 5.1 will help calculate the maximum dose for a specific agent depending on the weight of the patient. These dosages are averages, however, and the dentist must be alert to adverse systemic effects when injected dosages approach the recommended limits.[1]
- Local anesthetics have different durations of action for pulpal and soft-tissue anaesthesia. Pulpal (deep) anaesthesia varies *from 30 to 90 or more minutes*. Soft-tissue anaesthesia varies *from 1 to 9* h, depending on the specific agent and whether or not a vasoconstrictor is included.
- Local anesthetics are selected on the basis of the estimated length of the clinical procedure and the degree of anaesthesia required (Box 5.1). Two (or more) anesthetic agents can be administered when needed. The total dose of both anesthetics should not exceed the lower of the two maximum doses for the individual agents. Anesthetics also are available in amide and ester types.
- Hypersensitivity and allergic reactions in affected patients are much less frequent with the *amide type* of local anesthetics.[1]

Clinical Considerations

Patient Factors

Cardiovascular System

Before administering any drug, the condition of the cardiovascular system (heart and blood vessels) must be assessed. At minimum, blood pressure, heart rate and rhythm should be evaluated and recorded for all the patients.

Table 5.1 Maximum recommended dosages (MRDs) of local anesthetics

	Manufacturer's and FDA (MRD)		
Local anesthetic	**mg/kg**	**mg/lb**	**MRD, mg**
Articaine			
With vasoconstrictor	7.0	3.2	None listed
Bupivacaine			
With vasoconstrictor	None listed	None listed	90
Lidocaine			
With vasoconstrictor	7.0	3.2	500
Mepivacaine			
No vasoconstrictor	6.6	3.0	400
With vasoconstrictor	6.6	3.0	400
Prilocaine			
No vasoconstrictor	8.0	3.6	600
With vasoconstrictor	8.0	3.6	600

	Calculation of milligrams of local anesthetic per dental cartridge (1.8 ml cartridge)		
Local anesthetic	**Percent concentration**	**mg/ml**	**× 1.8 ml = mg/cartridge**
Articaine	4	40	72*
Bupivacaine	0.5	5	9
Lidocaine	2	20	36
Mepivacaine	2	20	36
	3	30	54
Prilocaine	4	40	72

MRD, maximum recommended dose.
*Cartridges of some drugs in the United States read, '1.7 ml each'. The actual volume of all local anesthetic cartridges is approximately 1.76 ml.
(From Malamed SF: Handbook of local anaesthesia*, ed 6, St. Louis, MO, 2013, Mosby.)*

Clinical Notes

- According to the latest guidelines, patients with stage 1 hypertension are good candidates for all dental procedures:
  - ***Systolic pressure less than 140 mmHg.***
  - ***Diastolic pressure less than 90 mmHg.***
- Patients with blood pressure consistently greater than the aforementioned numbers (stage 2 hypertension) should be referred to their physicians, particularly if the elevation is *greater* than 20 mmHg.[2]
- Malamed suggested that any resting patient with the following pulse rates be questioned further:
  - Pulse rate less than 60 beats per minute (beats/min).
  - Pulse rate greater than 110 beats/min.

Box 5.1 Approximate duration of action of local anesthetics*

Short duration (pulpal anaesthesia approximately 30 min)
Mepivacaine, HCl 3%
Prilocaine, HCl 4% (by infiltration)
Intermediate duration (pulpal anaesthesia approximately 60 min)
Articaine, HCl 4% + epinephrine 1:100,000
Articaine, HCl 4% + epinephrine 1:200,000
Lidocaine, HCl 2% + epinephrine 1:50,000
Lidocaine, HCl 2% + epinephrine 1:100,000
Mepivacaine, HCl 2% + levonordefrin 1:20,000
Mepivacaine, HCl 2% + epinephrine 1:200,000
Prilocaine, HCl 4% (via nerve block only)
Prilocaine, HCl 4% + epinephrine 1:200,000
Long duration (pulpal ≥90 min) anaesthesia approximately 90+ min
Bupivacaine 0.5% + epinephrine 1:200,000

*These anesthetics all are from the amide category.
(From Malamed SF: *Handbook of local anaesthesia*, ed 6, St. Louis, MO, 2013, Mosby.)

- Patients with valvular heart disease or a predisposition to bacterial endocarditis should have prophylactic antibiotics prescribed before dental treatment; the American Heart Association defines the recommended regimen for these antibiotics.[3]
- Overdose of any vasoconstrictor causes increased blood pressure, elevated heart rate and possible dysrhythmias.

Central Nervous and Respiratory Systems

The central nervous system (CNS) is more easily affected by overdose of injected anesthetic drugs than the cardiovascular system. Anesthetics depress the CNS, but when administered properly for local anaesthesia, little or no clinical evidence of depression is encountered.

- At *minimal to moderate* overdose levels, depression is manifested in excitation (e.g. talkativeness, apprehension, sweating, elevated blood pressure and heart rate, elevated respiratory rate) or drowsiness.
- At *moderate to high* overdose levels, tonic–clonic seizure activity may occur, followed by generalized CNS depression, depressed blood pressure, reduced heart rate (<60 beats/min), depressed respiratory rate and respiratory arrest.

Clinical Notes

With lidocaine and prilocaine, the usual progression of excitatory signs and symptoms described previously may not be seen, and the first clinical evidence of overdose may be *mild sedation or drowsiness*.[1]

Allergy

Malamed stated that documented, reproducible allergy is an absolute contraindication for administration of local anesthetics.[1] When a patient reports a history of '*sensitivity*' or '*reaction*' to an injected dental anesthetic, the dentist must believe the patient until further investigation disproves the patient's claim. Anaphylactic shock from an allergic reaction can be immediate and life threatening.

Any special condition of the patient should be recorded in the chart. The health status of the cardiovascular system, CNS, respiratory system, liver, kidneys and thyroid gland should be noted, as well as the patient's age, allergies and pregnancy status. A medical history form must be completed and signed by the patient.

Clinical Notes

- *Fast injection* and *intravascular injection* of anesthetics are the reasons for allergy-like reactions reported by patients.
- Some patients have proven allergy to *bisulphite*, an antioxidant used in anesthetic cartridges as a preservative for the vasoconstrictor.[1]

Benefits of Local Anesthesia

Cooperative Patient

- When a local anesthetics appropriate for the procedure is properly administered, patient anxiety and tension should be minimal.
- The appreciation and trust of the patient for the dentist is expressed in a more relaxed and cooperative attitude.
- Physically and *emotionally*, the patient and the dentist benefit from a relatively calm environment.

Salivation Control

Saliva control is a *primary reason* for the use of profound anaesthesia in most patients. For years, it has been observed that complete anaesthesia of all tissues (teeth and gingival tissues) in the dental operating site results in a marked reduction of salivation.[4]

Haemostasis

The term *haemostasis,* as used in operative dentistry, refers to the temporary reduction in blood flow and volume in tissue (ischaemia) where a vasoconstrictor is used. The alpha receptor effect of the vasoconstrictor causes constriction of the small blood vessels; the affected tissue bleeds less if cut or abraded. The principal function of a vasoconstrictor in operative dentistry is the prolongation of anaesthesia because of reduced blood flow to and from the anaesthetized site. Reduced blood flow helps keep the patient's blood level of the anesthetic and the vasoconstrictor at a low level by reducing the rate of absorption into the circulatory system.

Clinical Notes

- Without epinephrine, anaesthesia from 1 ml of lidocaine 2% lasts only 5–10 min.
- With epinephrine, the anaesthesia lasts 40–60 min.

Operator Efficiency

Local anaesthesia greatly benefits the dentist and the patient and is beneficial for successful tooth preparation and restoration. It improves the operator efficiency, and usually, the patient is calmer and more cooperative. This increased cooperation may reinforce the dentist's confidence and calmness, which may promote more efficient treatment. Without distractions or management problems from the patient, the dentist can focus on the treatment and its completion within a reasonable time frame.

Administration

Psychology

Patients have varying degrees of concern about receiving an intraoral injection. A concerted effort by the dentist and dental assistant is required to make the procedure more acceptable, and a *positive caring approach* is desirable with all patients during this phase of treatment.

The operator must use a kind, considerate and understanding approach. Every assurance should be made that the comfort of the patient is paramount and that the teeth and soft tissues will be treated with care. Such assurances, confidently and softly spoken, are welcomed during the administration of local anaesthesia. One example is, '*I may be taking longer than you expected, but we are giving the solution slowly to be kind to your tissues*'. Patients who feel secure (safe from pain and in caring hands) gratefully accept local anaesthesia.

Clinical Notes

- The greatest positive effect is achieved through a caring manner, rather than by what is said.
- Words such as *pain*, *sting*, *hurt* and *inject* should not be used because no matter what else is said, the patient will remember these potentially fear-invoking words.
- The art of tactfully keeping the syringe and needle from the view of the patient should be practiced.

Principles

- *Infiltration anaesthesia* involves a supraperiosteal or field block, where deposition is near the nerve ends in the operating site. This is the preferred mode of anaesthesia for maxillary teeth.

- *Conduction anaesthesia* involves a nerve block, where deposition is near a nerve trunk at a distance from the operating site. This is the preferred mode of anaesthesia for mandibular teeth.
- The routine supine position of the patient helps prevent vasodepressor syncope because it maintains blood supply and blood pressure to the brain. As a precaution, the upper torso should *never* be more than 10 degrees below the horizontal plane because this may cause respiratory distress secondary to the force of viscera against the diaphragm.
- The syringe must have an *aspirating* feature. When anesthetic is administered, aspiration is second in importance only to slow deposition of solution.
- Injection into infected tissue should be avoided because of the risk of spreading the infection. Also, the anesthetic becomes less effective because the infected tissue is acidic rather than basic. Alternative approaches such as nerve block should be used.

Disposable needle

It is mandatory that for each patient (appointment), the dental assistant selects a sheathed, capped, new disposable needle of the desired length and gauge. The sheathed needle comes sterile from the manufacturer. The needle remains sheathed except for setting and testing the syringe preparedness, until the moment of entry at the injection site. This helps prevent the accidental needle-stick injury, which among other things indicates needle replacement.

Clinical Notes

- For each patient appointment, using a new, sterile needle is mandatory.
- The needle must be sufficiently long that its full length is never out of sight (never completely within tissue). This means that in the unlikely event that a needle breaks at the hub junction, some of the needle is exposed for grasping and withdrawal.
- The *27-gauge*, *long needle* is recommended for mandibular conduction anaesthesia of inferior alveolar nerve.
- The *30-gauge*, *short needle* is recommended for infiltration anaesthesia of maxillary teeth.
- The *30-gauge*, *long needle* should be avoided for conduction anaesthesia of the inferior alveolar nerve as it may deviate during injection.

Anesthetic solution

- The weakest solution of anesthetics that will be effective should be used.
- *Lidocaine 2% with 1:100,000 epinephrine* is commonly used in operative dentistry and is generally recommended.
- 1 ml provides infiltration of anaesthesia for 40–60 min for anterior teeth. The addition of a vasoconstrictor to the anesthetic solution is necessary to prolong anesthesia by decreasing the rate of absorption of the anesthetic into blood. The vasoconstrictor may reduce the potential of anesthetic toxicity.
- As previously described, the vasoconstrictor in the anesthetic solution administered by infiltration is useful in reducing the occasional haemorrhage by producing slight, transient ischaemia of the cut or abraded soft tissue.
- *Periodic aspiration* during injection is important to ensure that the solution is not being injected into a blood vessel. If the tip of the needle is in the vessel, blood is aspirated into the cartridge, indicating the need to reposition the needle. For patient safety and comfort, periodic aspiration is as important as slow deposition of the anesthetic solution.

Topical anesthetic

The use of a topical anesthetic is recommended prior to the injection. Before needle entry, the mucosa at the injection site should be wiped free of debris and saliva with sterile gauze. After this, a *lidocaine topical anesthetic ointment* is applied for a minimum of 1–2 min to the selected entry spot (using a cotton-tipped swab, limiting the area of application to the swab dimension).

Clinical Notes

Painless injection is achieved by a combination of the following:
- Use of a topical anesthetic.
- Use of sharp needles.
- Slow deposition rate (approximately 60 s per cartridge).

Injection site (maxillary infiltration)

- With the left hand, a right-handed operator gently raises the lip outward and upward to identify the vestibular fornix, or the mucogingival junction, where the attached gingiva joins the alveolar mucosa (Fig. 5.1).
- Holding the lip high enough, the operator visualizes the location of the root end and determines the injection site in the alveolar mucosa.
  - As it is stretched perpendicular, or nearly so, to the long axis of the tooth and
  - Towards the periosteal target area that is very near the root end of the tooth to be operated on (Fig. 5.1).
- The injection site should be 5–10 mm lateral of the mucogingival line, allowing some freedom of needle movement and avoid causing tissue tension. If the needle is held parallel to the tooth long axis, rather than at an angle as recommended, the needle tends to enter too close to the attached mucosa and thus to the sensitive periosteal lining, which would cause pain from touching or stripping the periosteum from

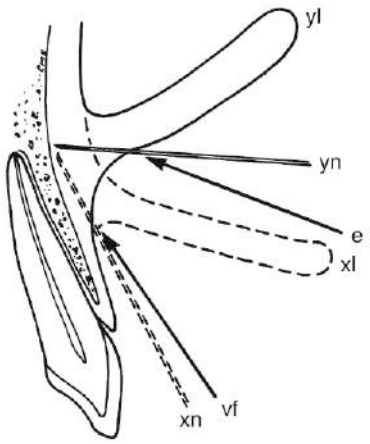


Figure 5.1 Recommended entry spot (*e*), direction of needle (*yn*) and lip position (*yl*) for infiltration anaesthesia of the maxillary canine. Direction of needle (*xn*) and lip position (*xl*) are not recommended. Vestibular fornix (*vf*) is the junction of the loose and fixed mucosa.

the bone. The needle tip should not be close to the periosteum until it has reached its target area.

Injection

Step 1: With the injection site identified and the needle directed properly, two things are done simultaneously in preparation for injection:

- A slight, gentle tug is applied to the lip (outward and upward) to have the entry spot tissue slightly taut and
- The needle is inserted about 3 mm into the mucosa (all the bevel under the epithelium).

The slight, gentle tug while tensing the tissue, coupled with topical anaesthesia, masks any sensation from the needle entry. After this, the lip may be relaxed, maintaining visibility of the needle.

Step 2: Initially, a small amount of solution is deposited slowly while observing and reassuring the patient. The operator waits several seconds for the anesthetic to take effect near the injection site before continuing the injection.

Step 3: Still maintaining proper needle direction, the operator gently continues inserting the needle towards the targeted periosteum. The operator needs to be careful to sense resistance when the needle tip touches the lining of the bone, at which time the needle immediately is withdrawn 1–2 mm.

Step 4: The operator aspirates by slightly reversing the plunger a few millimetres by gentle backward movement of the thumb ring. Aspiration (negative pressure) verifies that the needle is not in a blood vessel. If blood appears in the cartridge, the aspiration is positive; the needle is immediately withdrawn 1–2 mm, and the operator aspirates again until blood does not appear.

Step 5: If aspiration is negative, the operator slowly deposits 1 ml (slightly more than half the cartridge) over the next 30 s, continually observing and reassuring the patient. A rate of deposition of *1* min *for 1* ml is a good rule of thumb.

Clinical Notes

- *Slow deposition* is the most important safety procedure for the prevention of adverse reactions from high blood levels of the anesthetic or epinephrine.
- *Aspiration* is second in importance.
- Malamed defined overly rapid deposition as taking *less than 30* s *for 1.8* ml (one cartridge).[1] This fast rate separates tissue and is too rapid to allow diffusion along normal tissue planes. If the injection is intravascular, it can lead to serious adverse reactions. Also, it is painful or at least uncomfortable.
- Malamed stated that a *1*-min *rate for 1.8* ml *of anesthetic* (30 s for 1 ml, or half a cartridge) would not cause tissue damage and would not lead to serious overdose reactions, even if the anesthetic is accidentally injected intravascularly.[1]

Step 6: After deposition, the needle is gently withdrawn and resheathed.

Clinical Notes

- *Resheathing* is crucial in the prevention of needlestick injuries, which can cause cross-infection to the operator and other office personnel. The Occupational Safety and Health Administration (OSHA) stipulates that needle resheathing should be in a one-handed procedure.[5]
- It is also recommended that resheathing should be done by the *same person* who gave the injection; this eliminates the hazard of passing exposed needles. Even though multiple injections using the same needle for a patient creates no infection control concerns, multiple uses are discouraged because the used needle and the contents of its lumen may be infectious to dental personnel if accidental needlestick injury occurs.

Step 7: It is important that the patient be *continually observed* during and after the administration of local anaesthesia. An anaesthetized patient should never be left unattended and unobserved. Adverse reactions, if they occur, demand immediate attention by the dentist.

Disposal of needle and cartridge

Proper disposal of the needle and cartridge is crucial. Removal and disposal of the sheathed, used needle is done by the dental assistant. Tissue contact with the uncapped, exposed cartridge needle should be avoided. Disposal of the sheathed, used needle into a nearby sharps disposal container is recommended. The cartridge also should be disposed of in the sharps container. The sharps container

must be leak-proof and hard-walled and display an OSHA biohazard label.[5]

Emergency Procedures

The importance of taking pretreatment vital signs cannot be overemphasized. Adverse reactions occurring during or after administration of local anaesthesia can lead to serious complications that require emergency procedures. Foremost among these procedures are the following:

- Place the patient in a supine position (note the exception below).
- Summon medical assistance.
- Monitor vital signs.
- Apply basic life support (open the airway and use cardiopulmonary resuscitation [CPR], if needed).

Clinical Notes

- The supine position, with legs (only) slightly elevated, increases the volume of circulating blood and aids in increasing blood pressure. This procedure for a patient in *syncope*, or with syncopal symptoms, should relieve hypoxia of the brain and return the patient to, or help maintain, consciousness.
- The supine position should not be used, however, when symptoms (e.g. fear, perspiration, weakness, pallor, palpitations) suggest an *epinephrine overdose*. In this case, a semierect or sitting position is best because it minimizes any further elevation in cerebral blood pressure.[1]

Analgesia (Inhalation Sedation)

The most appropriate method of preventing pain is by blocking the nerve pathways capable of conducting nerve impulses. For patients who have a low threshold of pain and are apprehensive (hyper-responders), raising the threshold by inhalation sedation is an adjunctive aid to anaesthesia by injection. The use of *nitrous oxide and oxygen* is one method of inhalation sedation. The reader is referred to a textbook on anaesthesia that covers inhalation sedation in detail. The operator should understand that this method of pain control has definite limitations.

Clinical Notes

Analgesia should not be thought of as general anaesthesia in any stage or depth. It is simply a *condition in which the pain threshold is elevated*. With inhalation sedation, the patient is conscious of the activities around him or her.

Summary

Pain control forms the fundamental prerequisite for optimal standard of patient care. A sound knowledge of the clinical considerations, principles and clinical technique is mandatory for a clinician to achieve adequate pain control.

References

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